Finals

Lec - 01

Day :

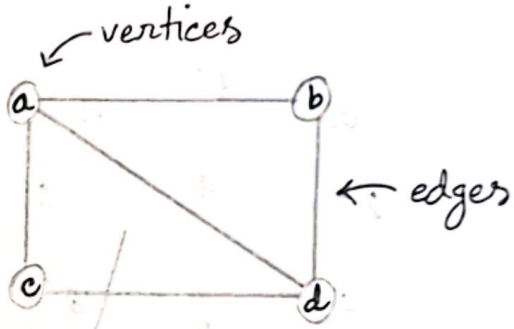
Time : Date : / /

What is a graph ?

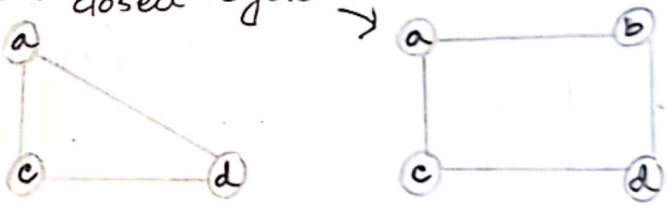
⇒ A Graph is a non-linear data structure that has -

i) Edges
ii) Vertices
iii) Closed Cycle

Example :-



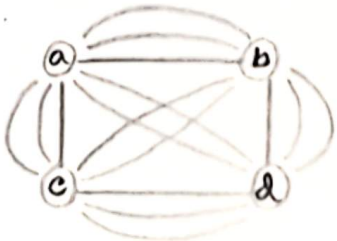
closed Cycle →



A graph G = (V, E)

V → Vertices , E → Edges

Dense graph : $|E| = |V|^2$; Sparse graph : $|E| = |V|$

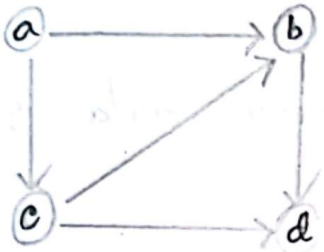




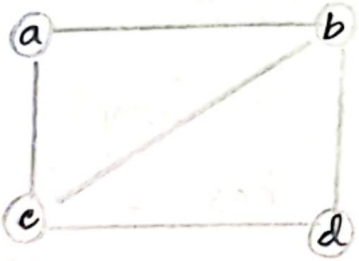
CamScanner

Day :

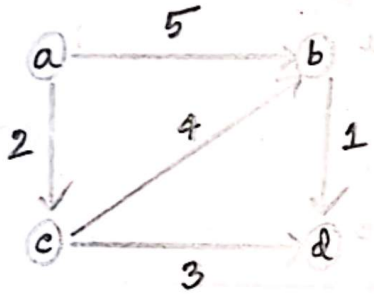
Time : Date : / /



(directed)
Graph



(Undirected)
Graph



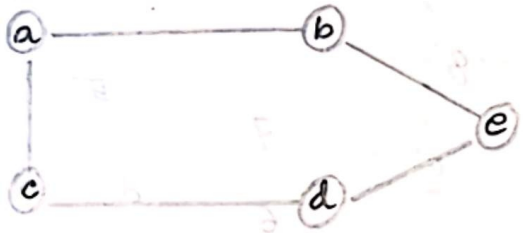
(Weighted + directed)
Graph

Day : _____

Time : Date : / /

Adjacency Matrix Representation

* For undirected & unweighted graph



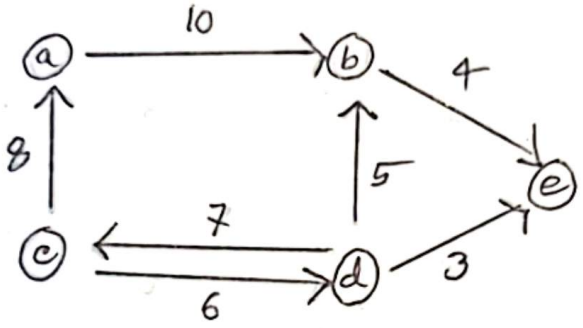
	a	b	c	d	e
a	0	1	1	0	0
b	1	0	0	1	1
c	1	0	0	1	0
d	0	1	1	0	1
e	0	1	0	1	0

CamScanner

Day : _____

Time : Date : / /

Adjacency Matrix Representation for directed and weighted graph



	a	b	c	d	e
a	0	10	0	0	0
b	0	0	0	0	4
c	8	0	0	6	0
d	0	5	7	0	3
e	0	0	0	0	0

CamScanner

Day :
Time : Date : / /

Graph Traversal

* What is Graph Traversal ?

⇒ Graph Traversal is the process of visiting all the nodes / vertices in a graph exactly once in a systematic way.

Two main approaches of Graph Traversal

1. Breadth - First Search (BFS)

Vertex colors used in BFS

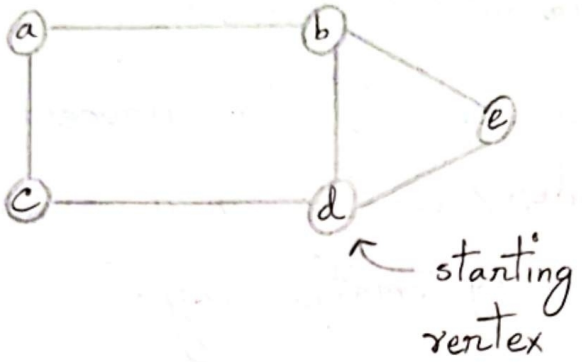
* White : Vertex not visited
* Grey : Vertex visited but not completed
* Black : Vertex visited and Completed.

CamScanner

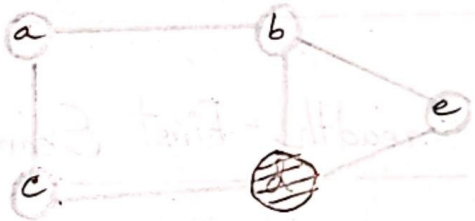
Day :

Time : Date : / /

Practice



step 01 :



Queue

↳ q :- d

step 02 :

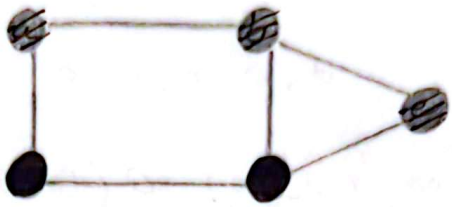


q : c, b, e

Day : _____

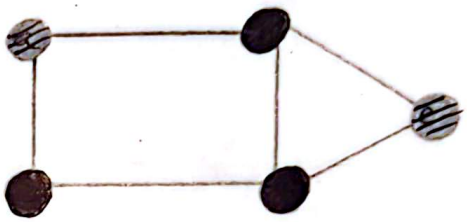
Time : Date : / /

step 03 :



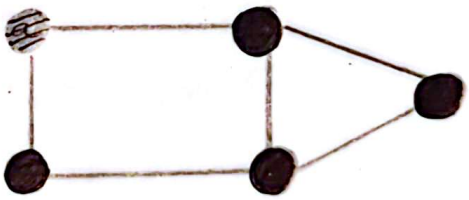
Q: b, e, a

step 04 :



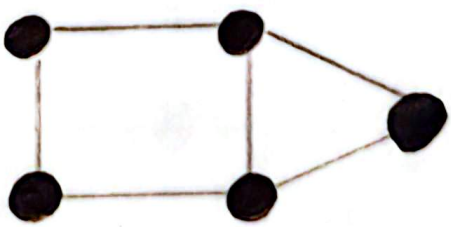
Q: e, a

step 05 :



Q: a

step 06 :



Q: $\emptyset$

CamScanner

Day :

Time : Date : /

Answer : d, c, b, e, a

Time Complexity : O(E+V)

Space Complexity : O(VxV)

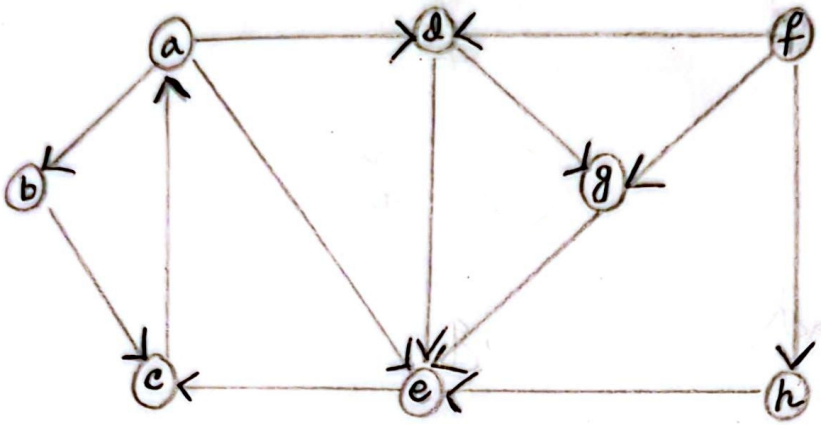
CamScanner

Lec-02

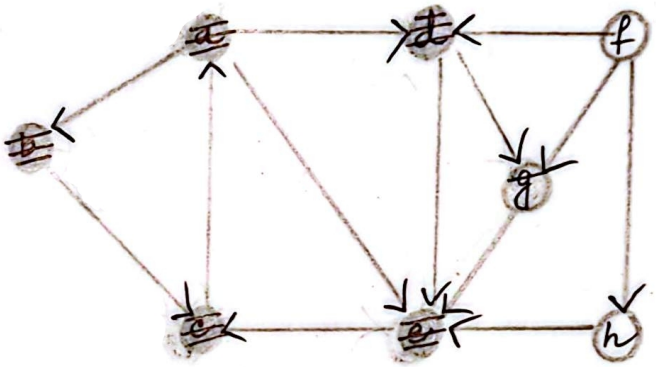
Day : _____

Time : Date : / /

(DFS) Depth First Search

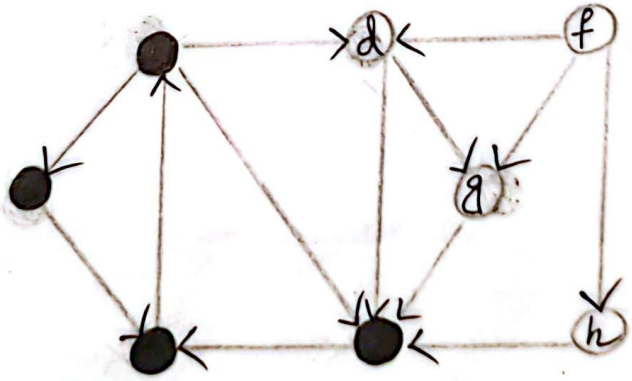


step 01 :



stack : d, e, c, a, b

step 02 :



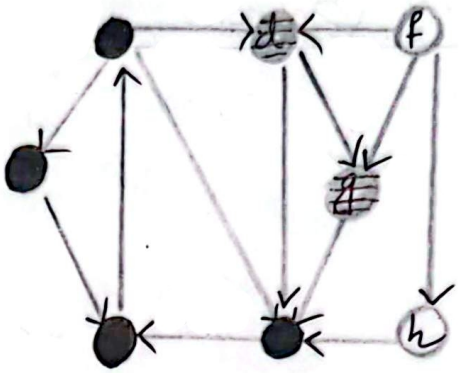
stack : d

CamScanner

Day : _____

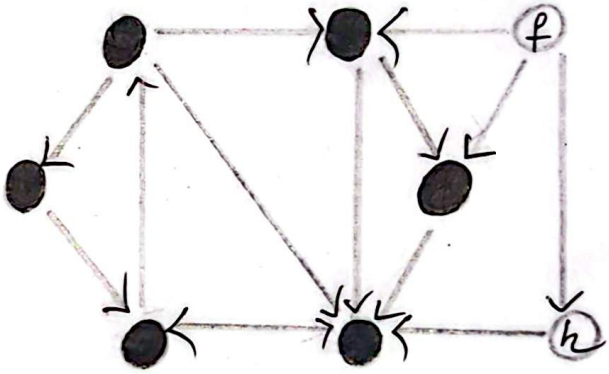
Time : Date : /

step 03 :



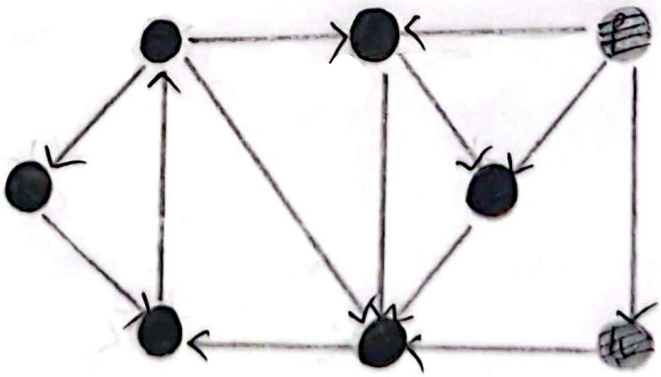
stack : ~~ø~~ d, g

step 04 :



stack : ø

step 05 :

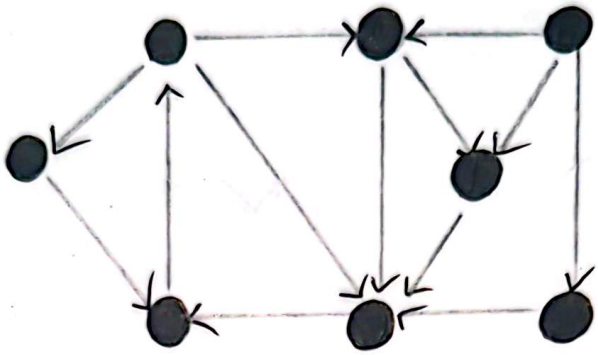


stack : f, h

Day : _____

Time : Date : / /

step 06 :



stack : ∅

Answer : d, e, c, a, b, g, f, h

Time Complexity : O(V+E)

CamScanner

Day : _____

Time : Date : / /

(MST) Minimum Spanning Tree

Input - Graph

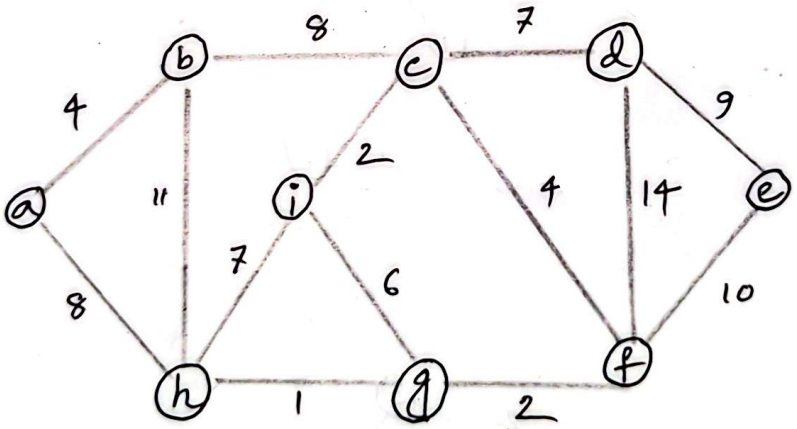
Output - Tree

→ Uses 2 algorithms

i) Kruskal's Algorithm (work's with edges)

ii) Prim's Algorithm (work's with vertexes)

Kruskal's Algorithm



Day :

Time : Date : /

(h,g) - 1 ~~(i,g)~~ -6 (d,e) -9

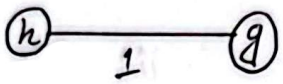
(g,f) - 2 (c,d) -7 ~~(e,f)~~ -10

(i,c) - 2 ~~(i,c)~~ -7 ~~(b,h)~~ - 11

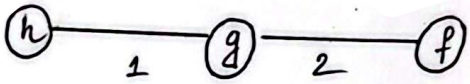
(c,f) - 4 (b,c) -8 ~~(d,f)~~ -14

(a,b) - 4 ~~(a,h)~~ -8

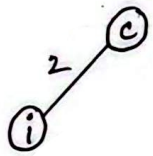
s1 :

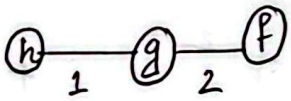


s2 :

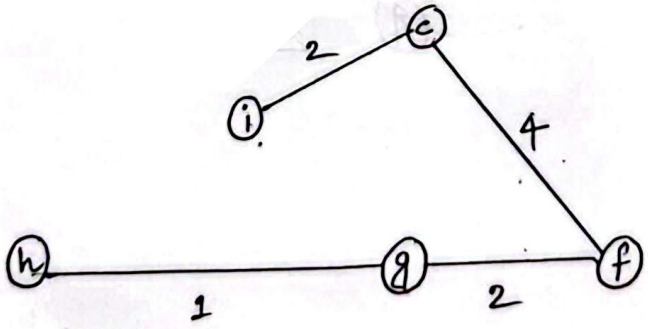


s3 :





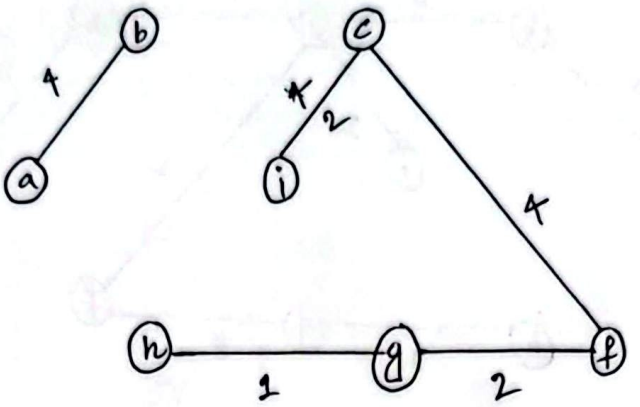
s4 :



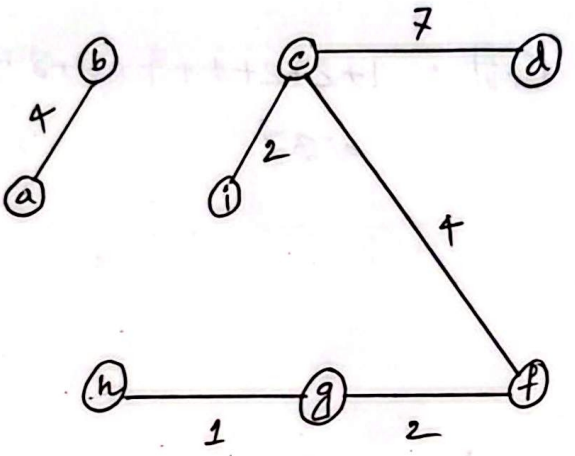
Day : _____

Time : Date : / /

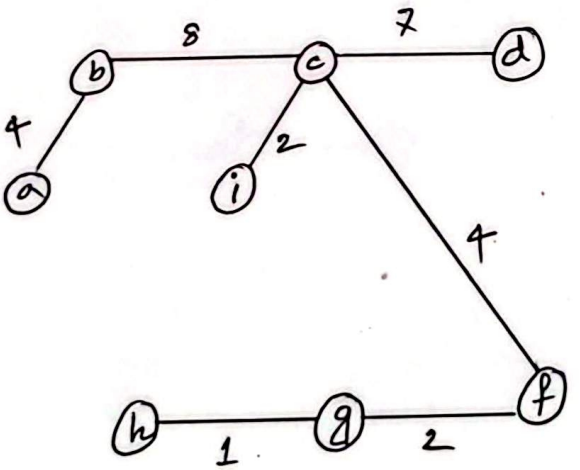
s5 :



s6 :



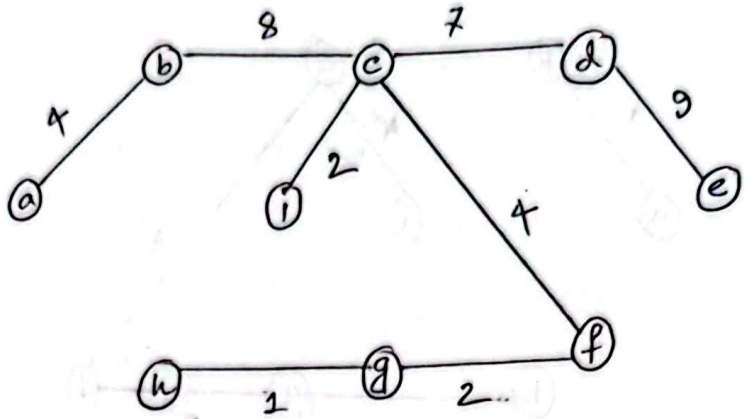
s7 :



Day :

Time : Date : /

s8 :



~~s9~~ :

Total Cost : 1+2+2+4+4+7+8+9

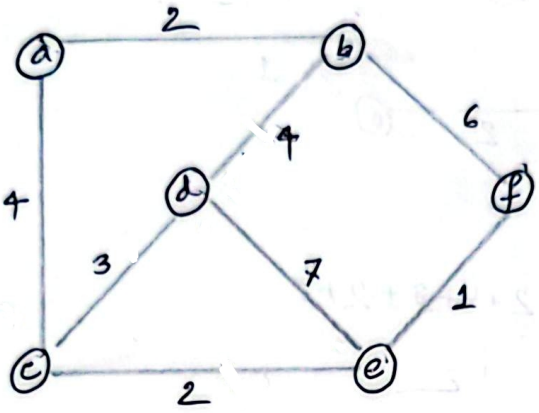
= 37

Day : _____

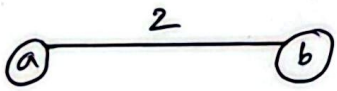
Time : Date : / /

Prim's Algorithm

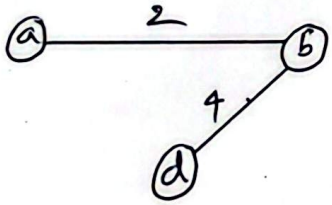
starting index →



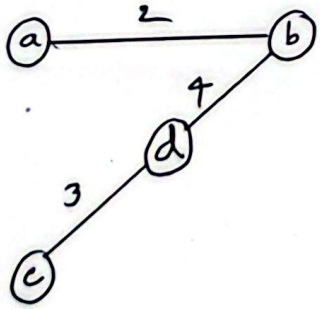
s1 :



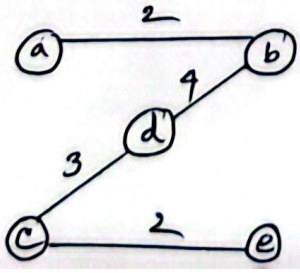
s2 :



s3 :



s4 :

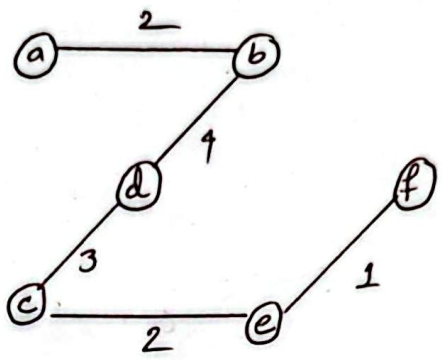


CamScanner

Day : _____

Time : Date :

s5 :



Total cost = 2+4+3+2+1

= 12

Day :

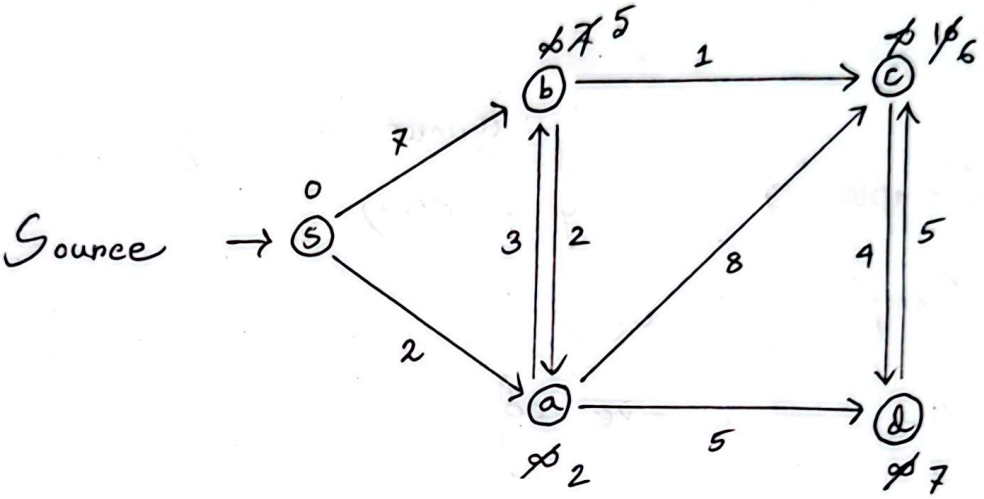
Time : Date : / /

Lec-03

Single Source Shortest Path

Dijkstra's Algorithm

Directed Graph



	s	a	b	c	d	
Initial	[0]	∞	∞	∞	∞	
s	0	[2]	7	∞	∞	
a	0	2	[5]	10	7	
b	0	2	5	[6]	7	
c	0	2	5	6	[7]	
d	0	2	5	6	7	

Day : _____

Time : Date : /

Total Path :- $s \longrightarrow a = 2$

$s \longrightarrow a \longrightarrow b = 5$

$s \longrightarrow a \longrightarrow b \longrightarrow c = 6$

$s \longrightarrow a \longrightarrow d = 7$

Time Complexity : Priority Queue $\rightarrow O(E \log V)$

Array $\rightarrow O(V^2)$

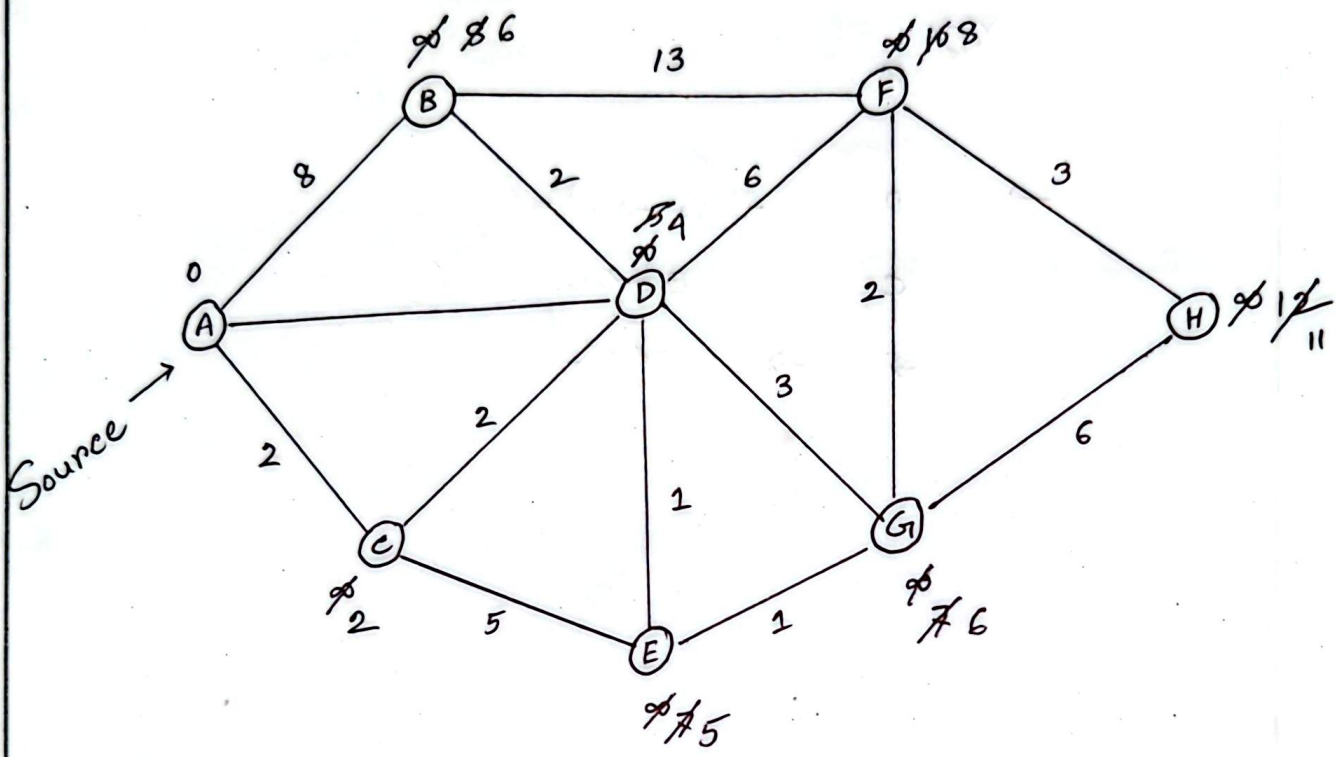
Space Complexity : $O(V)$

Limitation : -ve edge

Approach : Greedy

Day : _____

Time : Date : / /



	A	B	C	D	E	F	G	H
Initial	[0]	∞	∞	∞	∞	∞	∞	∞
A	0	8	[2]	5	∞	∞	∞	∞
C	0	8	2	[4]	7	∞	∞	∞
D	0	6	2	4	[5]	10	7	∞
E	0	[6]	2	4	5	10	6	∞
B	0	6	2	4	5	10	[6]	∞
G	0	6	2	4	5	[8]	6	12
F	0	6	2	4	5	8	6	[11]
H	0	6	2	4	5	8	6	11

Total Path : A → C → D → B = 6

A → C = 2

A → C → D = 4

A → C → D → E = 5

CamScanner

Day : _____

Time : Date :

A → C → D → E → G → F = 8

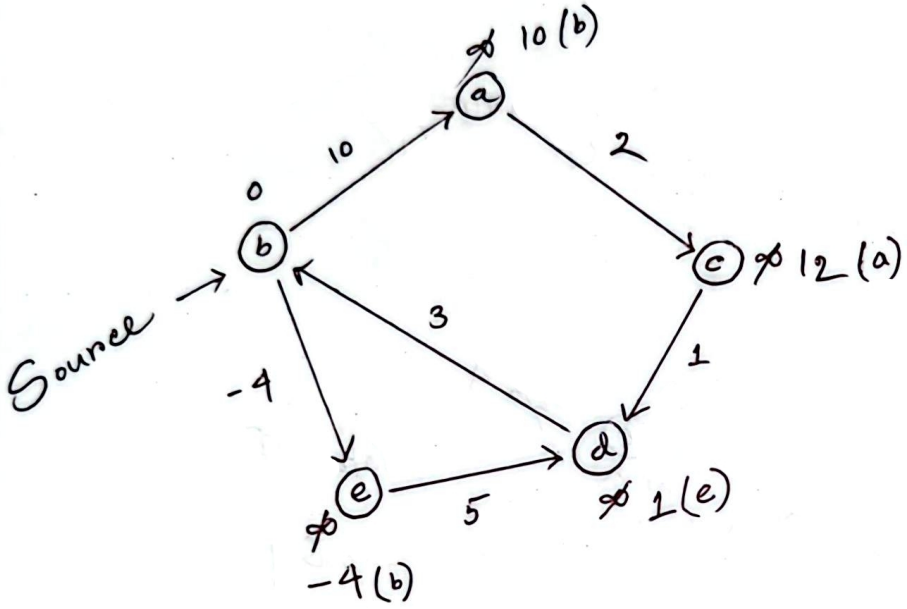
A → C → D → E → G = 6

A → C → D → E → G → F → H = 11

Day :

Lec-04 Time : Date : / /

Bellmanford - SSSP



(b,e)
(b,a)
(a,c)
(c,d)
(d,b)
(e,d)

steps = v-1

step	b	a	c	d	e
Initial	0	∞	∞	∞	∞
P_1	0	10 b	12 a	1 e	-4 b
P_2	0	10 b	12 a	1 e	-4 b
P_3	0	10 b	12 a	1 e	-4 b
P_4	0	10 b	12 a	1 e	-4 b

Total Path : $b \rightarrow a = 10$

$b \rightarrow a \rightarrow c = 12$

$b \rightarrow e \rightarrow d = 1$

$b \rightarrow e = -4$

CamScanner

Day :

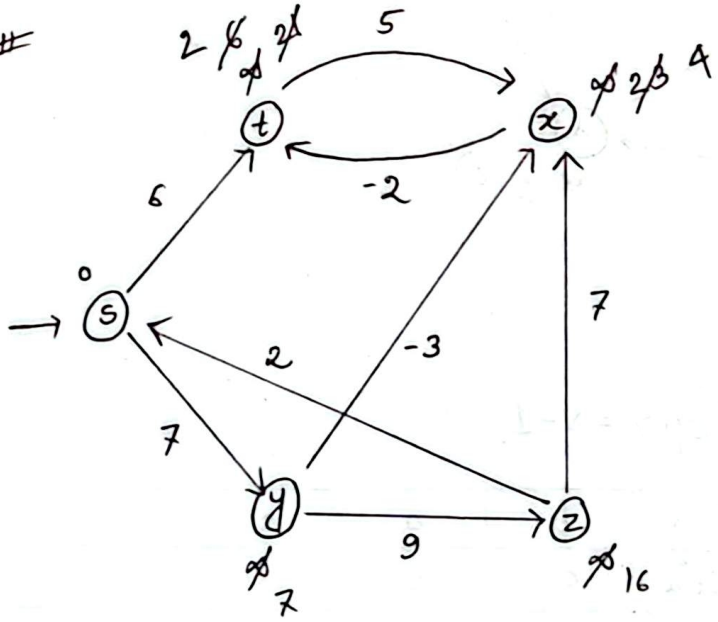
Time : Date : /

Time Complexity : $O(V \times E)$

Space Complexity : $O(V^2)$

Approach : Dynamic Programming

#



Total Path :-

$s \to t \to y \to x \to t = 2$

$s \to y \to x = 4$

$s \to y \to z = 16$

$s \to y = 7$

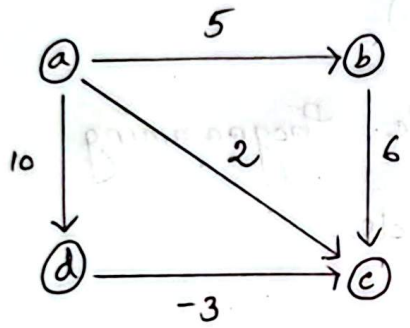
(s,y)
(y,z)
(z,x)
(x,t)
(t,x)
(s,t)
(y,x)
(z,s)

steps	s	t	x	z	y
Initial	0	∞	∞	∞	∞
P_1	0	6(s)	~~2~~ 4(y)	16(y)	7(s)
P_2	0	2(x)	4(y)	16(y)	7(s)
P_3	0	2(x)	4(y)	16(y)	7(s)
P_4	0	2(x)	4(y)	16(y)	7(s)

Floyd Warshall

Day :
Time : Date : / /

All pair shortest Path



$$D^0 = \begin{array}{c} \\ a \\ b \\ c \\ d \end{array} \begin{array}{c} \begin{array}{cccc} a & b & c & d \end{array} \\ \begin{bmatrix} 0 & 5 & 2 & 10 \\ \infty & 0 & 6 & \infty \\ \infty & \infty & 0 & \infty \\ \infty & \infty & -3 & 0 \end{bmatrix} \end{array}$$

$$D^a = \begin{array}{c} \\ a \\ b \\ c \\ d \end{array} \begin{array}{c} \begin{array}{cccc} a & b & c & d \end{array} \\ \begin{bmatrix} 0 & 5 & 2 & 10 \\ \infty & 0 & 6 & \infty \\ \infty & \infty & 0 & \infty \\ \infty & \infty & -3 & 0 \end{bmatrix} \end{array}$$

$$D^b = \begin{array}{c} \\ a \\ b \\ c \\ d \end{array} \begin{array}{c} \begin{array}{cccc} a & b & c & d \end{array} \\ \begin{bmatrix} 0 & 5 & 2 & 10 \\ \infty & 0 & 6 & \infty \\ \infty & \infty & 0 & \infty \\ \infty & \infty & -3 & 0 \end{bmatrix} \end{array}$$

$$D^c = \begin{array}{c} \\ a \\ b \\ c \\ d \end{array} \begin{array}{c} \begin{array}{cccc} a & b & c & d \end{array} \\ \begin{bmatrix} 0 & 5 & 2 & 10 \\ \infty & 0 & 6 & \infty \\ \infty & \infty & 0 & \infty \\ \infty & \infty & -3 & 0 \end{bmatrix} \end{array}$$

$$D^d = \begin{array}{c} \\ a \\ b \\ c \\ d \end{array} \begin{array}{c} \begin{array}{cccc} a & b & c & d \end{array} \\ \begin{bmatrix} 0 & 5 & 2 & 10 \\ \infty & 0 & 6 & \infty \\ \infty & \infty & 0 & \infty \\ \infty & \infty & -3 & 0 \end{bmatrix} \end{array}$$

CamScanner

Day :
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Time Complexity : $O(V^3)$

Space " : $O(V^2)$

Approach : Dynamic Programming

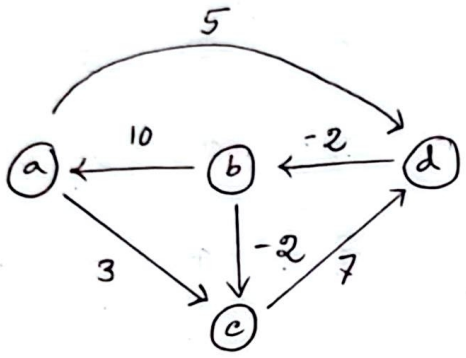
Limitation : -ve cycle

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Find all pair shortest path for the following graph



$D^0 =$

	a	b	c	d
a	0	∞	3	5
b	10	0	-2	∞
c	∞	∞	0	7
d	∞	-2	∞	0

$D^a =$

	a	b	c	d
a	0	∞	3	5
b	10	0	-2	15
c	∞	∞	0	7
d	∞	-2	-4	0

$D^b =$

	a	b	c	d
a	0	∞	1	5
b	10	0	-2	15
c	∞	∞	0	7
d	∞	-2	-4	0

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$$D^c = \begin{array}{c} \\ a \\ b \\ c \\ d \end{array} \begin{array}{c} \begin{array}{cccc} a & b & c & d \end{array} \\ \begin{bmatrix} 0 & \infty & 1 & 5 \\ 10 & 0 & -2 & 5 \\ \infty & \infty & 0 & 7 \\ \infty & -2 & -4 & 0 \end{bmatrix} \end{array}$$

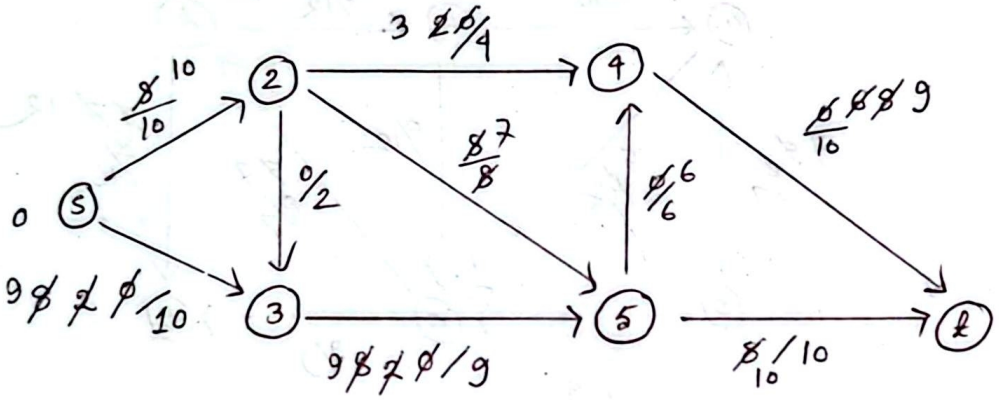
$$D^d = \begin{array}{c} \\ a \\ b \\ c \\ d \end{array} \begin{array}{c} \begin{array}{cccc} a & b & c & d \end{array} \\ \begin{bmatrix} 0 & 3 & 1 & 5 \\ 10 & 0 & -2 & 5 \\ 15 & 5 & 0 & 7 \\ \infty & -2 & -4 & 0 \end{bmatrix} \end{array}$$

CamScanner

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Ford Fulkerson : Maximum Flow



flow = 0 + 8 + 2 + 6 + 2 + 1
= 19

s → Source

t → Tank

Augmented Path	Minimum Cut
s → 2 → 5 → t	8
s → 3 → 5 → t	2
s → 3 → 5 → 4 → t	6
s → 2 → 4 → t	2
s → 3 → 5 → 2 → 4 → t	1

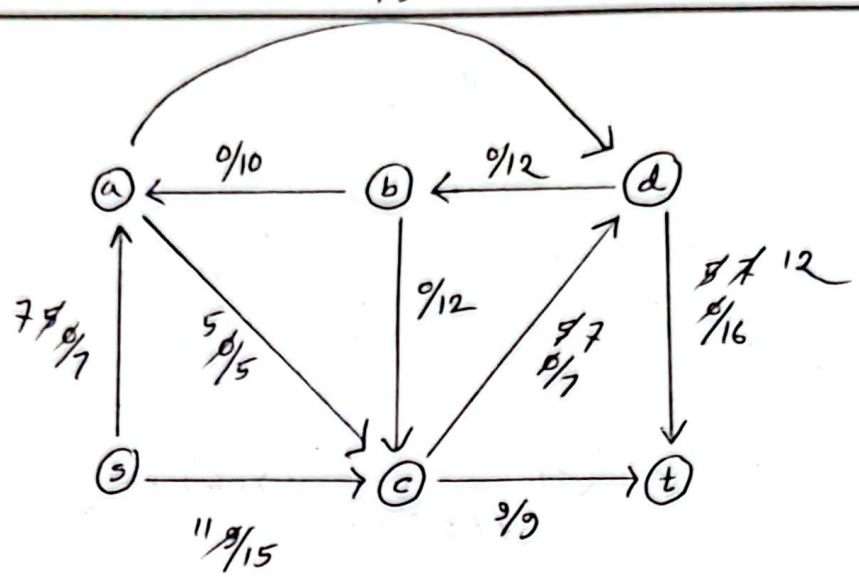
Time : $O(VXE^2)$ Limitation : -ve edge

Space : $O(V+E)$

Approach : DP

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flow = 9+5+2+2+5

=

Augmented Path	Minimum Cut
s → c → t	9
s → a → c → d → t	5
s → c → d → t	2
s → a → c → t	2
s → a → d → t	5

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Maximum - Sum Subarray

Kadane's Algorithm

A = 10, 7, 3, 1, 2, 4

Maximum Sum Subarray = 10+7+3+1+2+4

A = {-1}, -2, -3, -4, -6

Maximum Sum Subarray = -1

Example 01 : 6 -3 -2 3 -1 2

Cur Sum	Max Sum	
6	6	6 -3 -2 3 -1 2 (6)
3	6	6 -3 -2 3 -1 2 (3)
1	6	6 -3 -2 3 -1 2 (1)
4	6	6 -3 -2 3 -1 2 (4)
3	6	6 -3 -2 3 -1 2 (3)
5	6	6 -3 -2 3 -1 2 (5)

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Example 02 :

	2	-6	-1	3	-1	2	-2
CurSum							
2	2	2	-6	-1	3	-1	2	-2
-4	2	2	-6	-1	3	-1	2	-2
-1	2	2	-6	-1	3	-1	2	-2
3	3	2	-6	-1	3	-1	2	-2
2	3	2	-6	-1	3	-1	2	-2
4	4	2	-6	-1	3	-1	2	-2
2	4	2	-6	-1	3	-1	2	-2

Maximum Subarray = {3, -1, 2}

Max Sum = 4

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Maximum Subarray = {4, 1, 2}

Max Sum = 7

Time Complexity : O(n)

Space Complexity : O(n) / O(1)

Approach : DP

Recursion Relations

Recurrence
- Recursion Tree
- Substitution Method
- Master Theorem

Recursive
- Decreasing
- Dividing

Time Complexity : O(n)

```
void Test (int n) {
    if (n>0){
    for (i=0; i<n; i++){
      cout<<n;
    }
    } Test (n-1);
}
```

$T(n) = T(n-1) + n$

$T(n) - n$

$T(n-1) - (n-1)$

$T(n-2) - (n-2)$

$\vdots$

$T(1) \longrightarrow 1$

$\sum = 1+2+3+\cdots(n-2)(n-1)+n+$

$= \frac{n(n+1)}{2}$

$= \frac{n^2+n}{2}$

$= \frac{n^2+n}{2} \approx O(n^2)$

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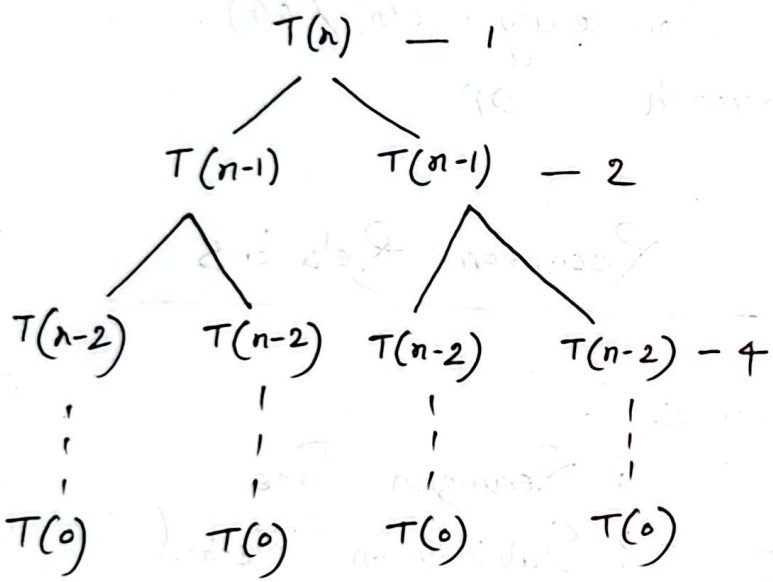
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Lec-07

Recursion Tree method

Ending
decreasing T(0)
dividing T(1)

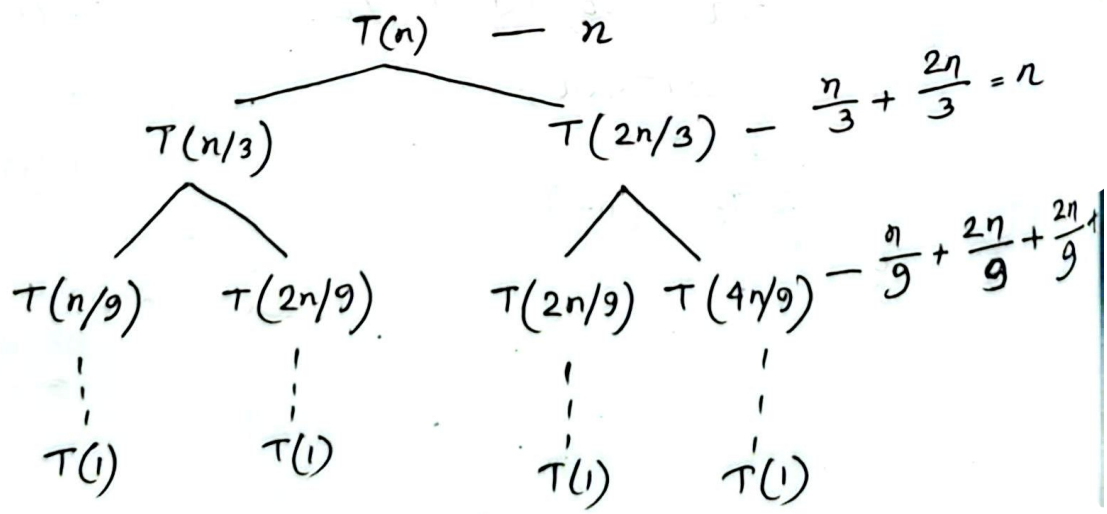
$T(n) = 2T(n-1) + 1$



$$1 + 2^0 + 2^1 + 2^2 + \cdots + 2^{(n)}$$

$$T.C \approx O(2^n)$$

$T(n) = T(n/3) + (2n/3) + n$



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Substitution Method

$T(n) = T(n-1) + 1$

$T(n-1) = T(n-1-1)+1$

$= T(n-2)+1$

$\Rightarrow T(n) = [T(n-2)+1]+1$

$= T(n-2)+2$

$T(n-2) = T(n-2-1)+1$

$= T(n-3)+1$

$\Rightarrow T(n) = [T(n-3)+1]+2$

$= T(n-3)+3$

$\vdots$

$\Rightarrow T(0) = T(n-k)+k$

Here, $n-k = 0$

$\Rightarrow n = k$

$\therefore T(n-n)+n$

$= T(0)+n$

$= 1+n$

T.e $= O(n)$

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$T(n) = T(n-1) + \log n$

$$T(n-1) = T\{(n-1)-1\} + \log(n-1)$$
$$= T(n-2) + \log(n-1)$$

$$\Rightarrow T(n) = T(n-2) + \log(n-1) + \log n \quad - (2)$$

$$T(n-2) = T(n-2-1) + \log(n-2)$$
$$= T(n-3) + \log(n-2)$$

$$\Rightarrow T(n) = T(n-3) + \log(n-2) + \log(n-1) + \log n$$

$$\vdots$$

$$\Rightarrow T(K) = T(n-K) + \log((n-(K-1)) + \log(n-(K-2)) \cdots + \log(n)$$

$$\therefore T(n-K) = T(0)$$

$$\Rightarrow n-K = 0$$

$$\Rightarrow K = n$$

$$\therefore T(n) = T(n-n) + \log((n-(n-1)) + \log(n-(n-2)) + \log(n-n\cdots-) + \log(n)$$

$$\log 1 + \log 2 + \log 3 + \cdots \log n$$

$$= \log(1 \times 2 \times 3 \times \cdots \cdots n)$$

$$= \log(n!)$$

$$= \log n^n = O(n \log n)$$